MILITARY SPECIFICATION

SCALES, GRAPHICAL FIRING: FAN TYPE, WITH EQUIPMENT
(FOR 81-MM AND 4.2 IN. MORTARS)

1. SCOPE

1.1 This specification covers two fan-type graphical firing scales used for determining and plotting data pertinent to indirect firing of mortars. Except where otherwise indicated, all statements pertinent to the graphical firing scales within this specification are applicable to both firing scales. The specification also covers the following items of equipment related to the scales: Case, Scale, Graphical Firing, Fan Type: 10527367; Case (Pin): 8578039; and Pin Assembly: 8578028. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

A10527360	Scale, Graphical Firing: Fan Type (for 81-mm mortar), W/E

FSC 1220

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DRAWINGS (Cont'd)

A10527361	Scale, Graphical Firing: Fan Type (for 4.2 in. mortar), W/E
B8578028	Pin Assembly: 8578028
C8578039	Case (pin): 8578039
F8578050	Scale, Graphical Firing: Fan Type (for 81-mm mortar)
F8578051	Scale, Graphical Firing: Fan Type (for 4.2 in. mortar)
F10527367	Case, Scale, Graphical Firing, Fan Type: 10527367

PACKAGING DATA SHEETS

MIL-P-14232/P10527360	Packaging of Scale, Graphical Firing: Fan Type (for 81-mm mortar), W/E
MIL-P-14232/P10527361	Packaging of Scale, Graphical Firing: Fan Type (for 4.2 in. mortar), W/E

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication of Graphical Firing Scales, Fan Type, W/E.

3.1.1 Graphical Firing Scale, Fan Type (for 81-mm mortar), W/E.- This firing scale, complete with equipment, shall be manufactured in accordance with Drawing A10527360 and all drawings pertaining thereto.

3.1.2 Graphical Firing Scale, Fan Type (for 4.2 in. mortar, W.E/. This firing scale, complete with equipment, shall be manufactured in accordance with Drawing A10527361 and all drawings pertaining thereto.

3.2 First article testing.- Requirements for the submission of first article samples of the graphical firing scale and items of equipment by the contractor for testing shall be as specified in the contract.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship

3.4 Graphical firing scales, Fan Type.

3.4.1 Fabrication.

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3.4.1.1 Graphical Firing Scale, Fan Type (for 81-mm mortar). - The firing scale shall be manufactured in accordance with Drawing F8578050 and all drawings pertaining thereto.

3.4.1.2 Graphical Firing Scale, Fan Type (for 4.2 in. mortar).- The firing scale shall be manufactured in accordance with Drawing F8578051 and all drawings pertaining thereto.

3.4.2 Environmental service conditions.

3.4.2.1 Storage temperatures.- The graphical firing scale shall show no evidence of damage or physical defects after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160° Fahrenheit (F).

3.4.2.2 Standard ambient temperature.- Subsequent to the requirements of 3.4.2.1 the graphical firing scale shall meet the requirements of 3.4.3 to 3.4.6 inclusive at a standard ambient temperature between +60° and +90°F.

3.4.2.3 Extreme operating temperatures.- The graphical firing scale shall meet the requirements of 3,4.6 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.4.3 Surface quality.- Surfaces of the graphical firing scale shall be free from stains and other imperfections.

3.4.3.1 Cursor.- The top frosted surface of the cursor shall be capable of accepting clear and legible markings from a lead pencil of 2H hardness. Upon erasure with a commercial pencil eraser, there shall be no evidence of permanent marks, smudges, *or* smears.

3.4.4 Inscribed data.- Graduation lines, letters, numerals, and other markings on the scales and the lines on the cursor shall be sharp, evenly colored, and without skips or breaks. Letters and numerals shall not touch each other or any line of any scale.

3.4.5 Accuracy.- Graduation lines shall be accurately located in accordance with the dimensions and tolerances shown on pertinent drawings.

3.4.6 Operability. - The fan scale, ballistic scales, and cursor shall assemble freely, and the cursor shall operate smoothly with the ballistic scales while the graphical firing scale is exposed to and thermally stabilized at the standard ambient temperature of 3.4.2.2 and the extreme operating temperatures of 3.4.2.3.

3.5 Case, Scale, Graphical Firing, Fan Type: 10527367.- The case shall be manufactured in accordance with Drawing F10527367 and all drawings pertaining thereto and requirements noted thereon,

3.6 Case (pin): 8578039.- The case shall be manufactured in accordance with Drawing C8578039 and requirements noted thereon.

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3.7 Pin Assembly - 8578028.- The pin assembly shall be manufactured in accordance with Drawing B8578028 and all drawings pertaining thereto.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4, and 4.6 as well as any additional tests that may be specified in the contract,

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4, and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot information and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

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4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I, II, and III herein. Examinations and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Surface quality	3.4.3	4.6.4
102. Cursor	3.4.3.1	4.6.5
103. Inscribed data	3.4.4	4.6.6
104. Accuracy	3.4.5	4.6.7
105. Operability	3.4.6	4.6.8
MINOR: NONE		

NOTE : The tests in Table I shall be conducted at temperatures between plus 600 and 90°F.

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One Graphical Firing Scale shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1, 3.4, 3.5, 3.6, 3.7	Applicable drawings- visual
302. General specification	3.3	MIL-F-13926 - visual

4.4.2 Environmental.- Three Graphical Firing Scales shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

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TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Extreme operating temperatures	3.4.2.3	4.6.2
305.	Storage temperatures	3.4.2.1	4.6.3

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test Methods and Procedures.

4.6.1 Fabrication. - The graphical firing scale and equipment as specified in the contract or purchase order shall be subjected to a visual, tactile, and dimensional inspection for conformance to the applicable drawings. Standard measuring equipment shall be used in performing dimensional inspection. The graphical firing scale and equipment shall conform to the applicable requirements of 3.1, 3.4, 3.5, 3,6, and 3.7.

4.6.2 Operating temperatures.- The test shall be in accordance with the "Test Facilities" requirements of Specification MIL-F-13926 and the conditions of 3.4.2.3. While thermally stabilized at -40° and +150°F, respectively, the assembly and operability of the fan scale, ballistic scales, and cursors shall be conducted by means of a visual and tactile examination. The components of the graphical firing scale shall conform to the requirements of 3.4.6.

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4.6.3 Storage temperatures.- Test equipment shall be in accordance with the "Test Facilities" requirements of Specification MIL-F-13926 and the conditions in 3.4.2.1. During temperature cycling, the fan scale, ballistic scales and cursor (that comprise a graphical firing scale) shall lay on a flat surface in the test chamber. The flat surface shall be capable of supporting the entire length and width of each component. Upon return to standard ambient temperature, the fan scale, ballistic scales, and cursor shall be subjected to a visual examination for evidence of damage or physical defects such as color change, separations, bubbles, tears and cracks. The graphical firing scale shall conform to the requirements of 3.4.2.1. The graphical firing scale shall then be subjected to the tests in 4.6.4 through 4.6.8 inclusive, and shall conform to the requirements specified.

4.6.4 Surface quality.- The surface quality of the graphical firing scale shall be visually inspected with the aid of a magnifier of at least 3 power (3X). The surface quality of graphical firing scales shall conform to the requirements of 3.4.3.

4.6.5 Cursor.- Perform this test with a commercial type quality lead pencil of 2H hardness. The pencil shall be equipped with a conical writing point and a clean commercial eraser. Lead pencil markings in the form of numerals and letters shall be placed throughout the frosted area of the cursor. All numerals and letters shall be clear, legible and without evidence of "skip". The frosted area of the cursor shall conform to the requirements of 3.4.3.1 when cleaned with the pencil eraser.

4.6.6 Inscribed data.- The lines, letters, numerals and other markings on the scales and cursor shall be visually inspected with the aid of a magnifier of at least 3 power (3X). The scales and cursor shall conform to the requirements of 3.4.4.

4.6.7 Accuracy.- The accuracy of the graduation lines shall be verified by comparison to a calibrated template. The graduation lines shall conform to the requirements of 3,4.5.

4.6.8 Operability.- The assembly and operability of the fan scale, ballistic scales, and cursor shall be conducted by means of a visual and tactile examination. The components of the graphical firing scale shall conform to the requirements of 3.4.6

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing, and marking shall be in accordance with MIL-P-14232 and the following packaging data sheets:

MIL-P-14232/P10527360	Packaging of Scale, Graphical Firing: Fan Type (for 81-mm mortar), W/E
MIL-P-14232/P10527361	Packaging of Scale, Graphical Firing: Fan Type (for 4.2 in. mortar), W/E

The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.

6.1.1 Scale, Graphical Firing; Fan Type (for 81-mm mortar).- The scale is used to determine firing data and for measuring angles and distances in conjunction with charts. or maps associated with indirect fire of 81-mm Mortar: M29,

6.1.2 Scale, Graphical Firing: Fan Type (for 4.2 in. mortar).- The scale is used to determine firing data and for measuring angles and distances in conjunction with charts or maps associated with indirect fire of 4.2 in. Mortar: M30.

6.1.3 Case, Scale, Graphical Firing, Fan Type: 10527367.- The case is used for stowing the graphical firing scale and Case (pin): 8578039 with Pin Assemblies: 8578028.

6.1.4 Case (pin): 8578039.- The case is used to stow Pin Assemblies: 8578028.

6.1.5 Pin Assembly: 8578028.- The pin assembly is used to establish a point on a map or chart about which the vertex of the fan scale pivots.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet number (see 5.1).
(c) Selection of applicable levels of preservation, packaging, and packing.
(d) Applicable stock number.
(e) Order of document precedence.

Custodian:

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Preparing activity:

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Project No. 1220-A244Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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